Formal Semantics

Concepts of Programming Languages

CAS CS 320

Outline

- Discuss **formal semantics** in general
- Look at **small-step** and **big-step** semantics with some examples

Introduction

A Thought Experiment

```bash
x=3
function f () {
    x=2
}
f
echo $x
```

Bash

```python
x = 3
def f():
    x = 2
f()
print(x)
```

Python

```ocaml
let x = 3
let f () =
  let x = 2 in
  ()
let _ = f ()
let _ = print_int x
```

OCaml

A Thought Experiment

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Question. *How do we know what'll be printed?*

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Question. *How do we know what'll be printed?*

As programmers, we build intuitions by writing programs

As PL designers, we have to *decide* this

Meaning

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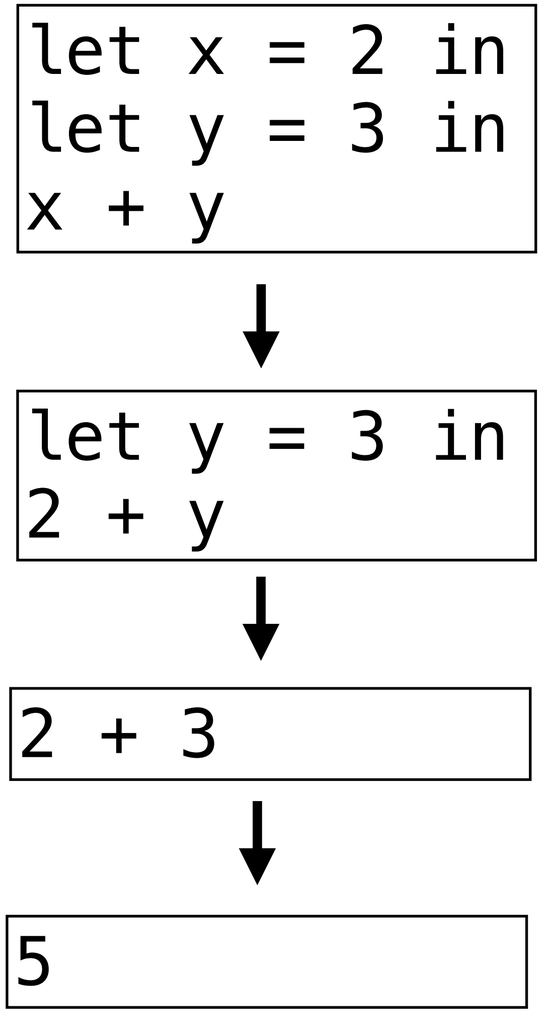
Syntax is interested in the *form* of a program

```
let x = 2 in
let y = 3 in
x + y
```

Meaning

Syntax is interested in the *form* of a program

Semantics is interested in the *meaning* of a program



Aside: Denotational vs. Operational Semantics

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Denotational semantics is interested in what a syntactic object "denotes" i.e. in interpreting programs as *mathematical objects*

$$1 + 2 * 3 + 4 = 11$$
$$1 + 12 - 2 = 11$$

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Operational semantics is interested in how a PL "operates" i.e. how a program *behaves* during execution

$$\begin{aligned} 1 + 2 * 3 + 4 &\longrightarrow 1 + 6 + 4 \\ &\longrightarrow 7 + 4 \\ &\longrightarrow 11 \end{aligned}$$

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This course

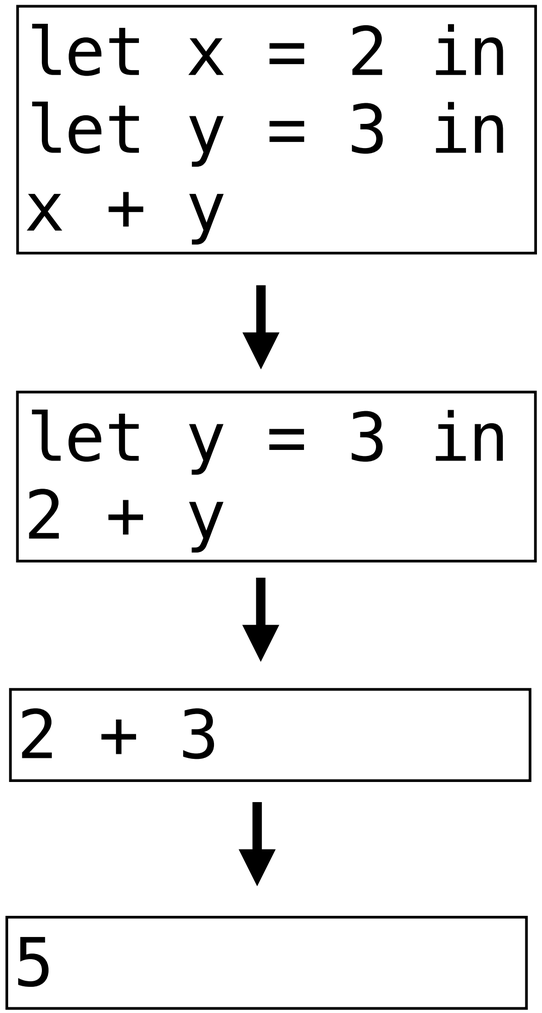
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$$\dfrac{\dfrac{2 \Downarrow 2 \qquad \dfrac{3 \Downarrow 3 \qquad 2 + 3 \Downarrow 5}{\texttt{let y = 3 in 2 + y} \Downarrow 5}}{\texttt{let x = 2 in let y = 3 in x + y} \Downarrow 5}$$

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320Caml

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Aside: Static vs. Dynamic Semantics

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Static semantics
refers to the meaning
given to a program
before it is evaluated

```
% ocaml silly.ml
File "./silly.ml", line 1, characters 8-9:
1 | let x = 2 +. 3.
            ^
Error: This expression has type int but an expression was
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  Hint: Did you mean `2.'?
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utop # let x = 2 + 3;;
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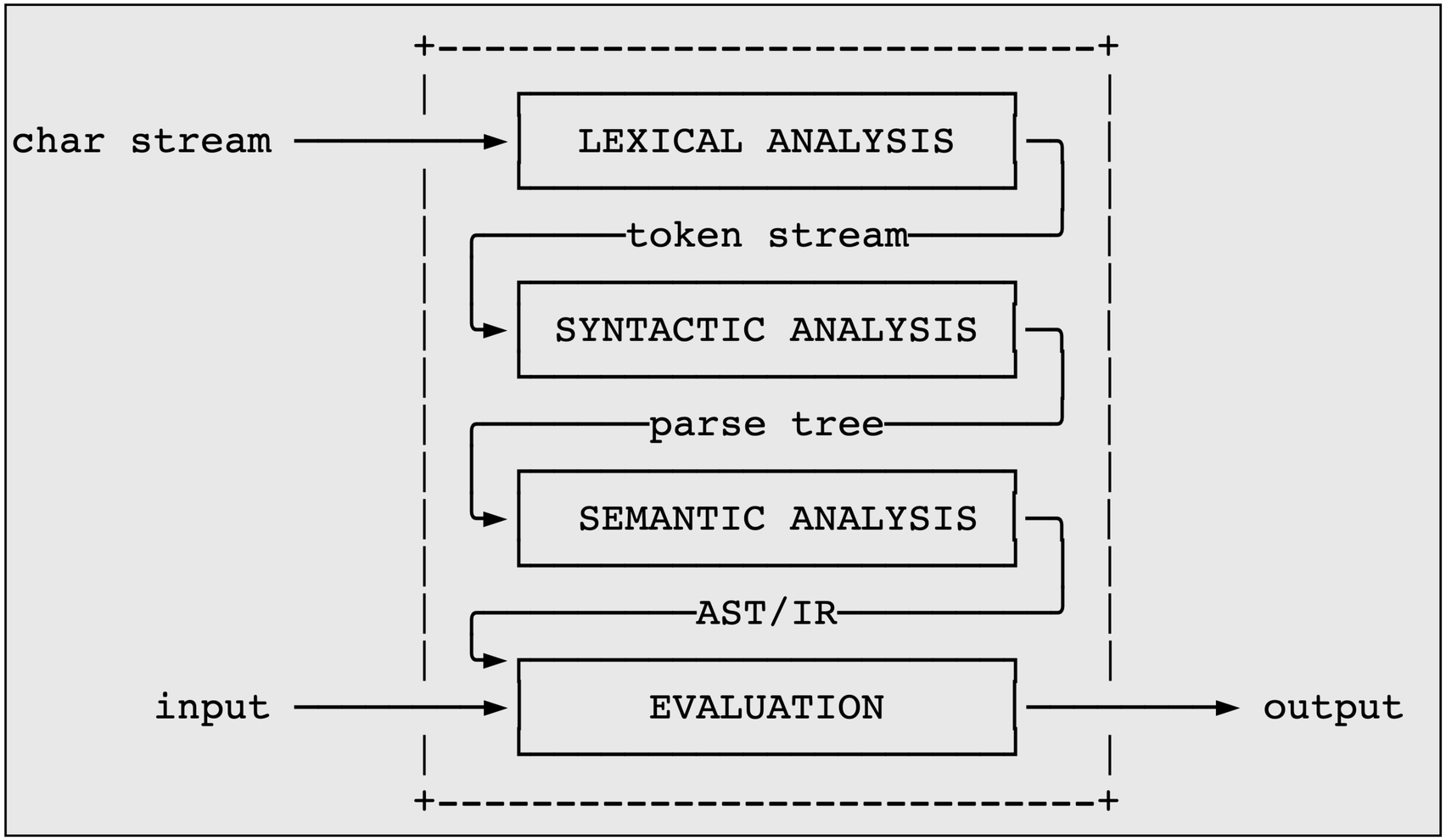
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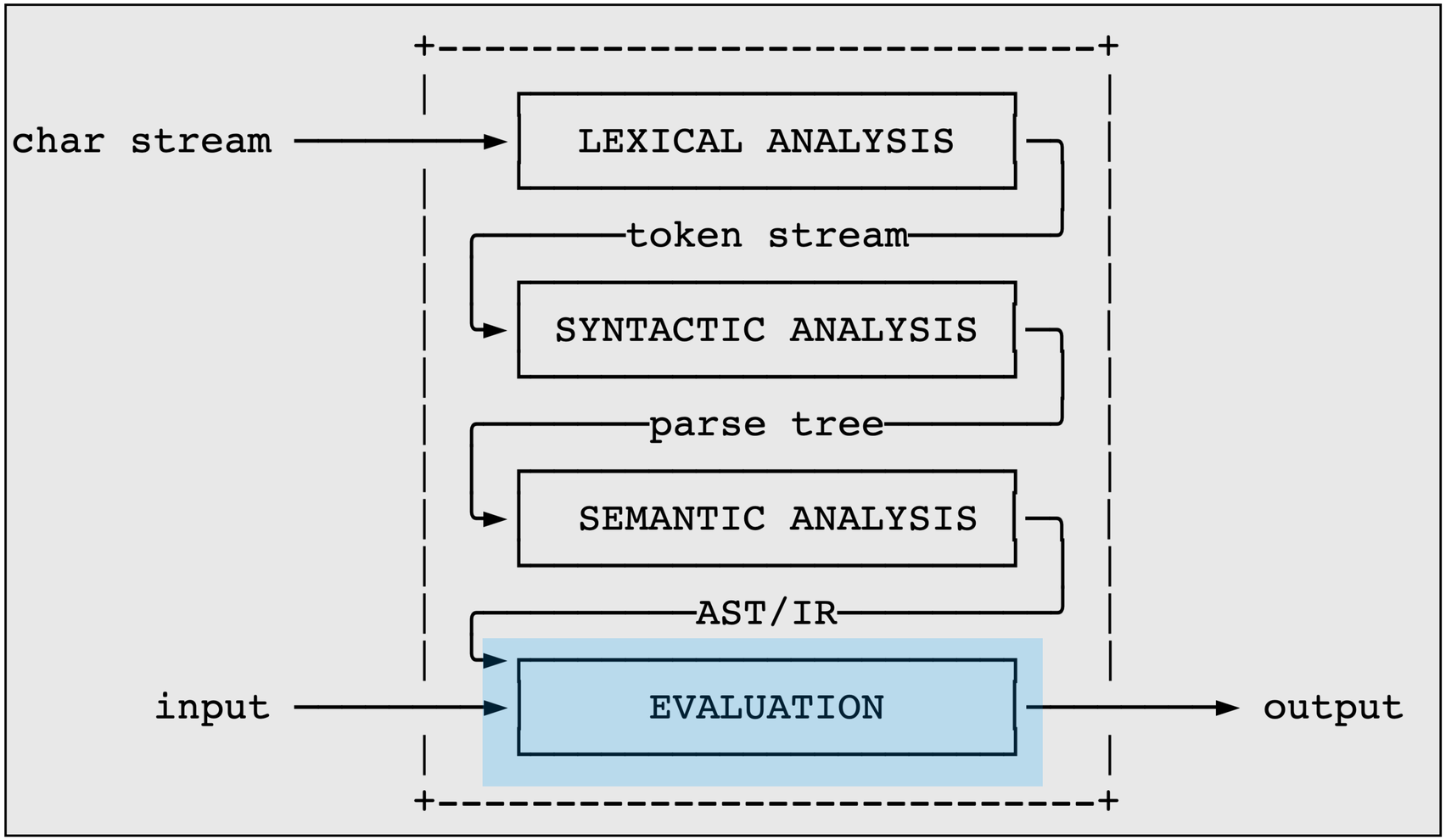
Evaluation

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Recall: The Picture

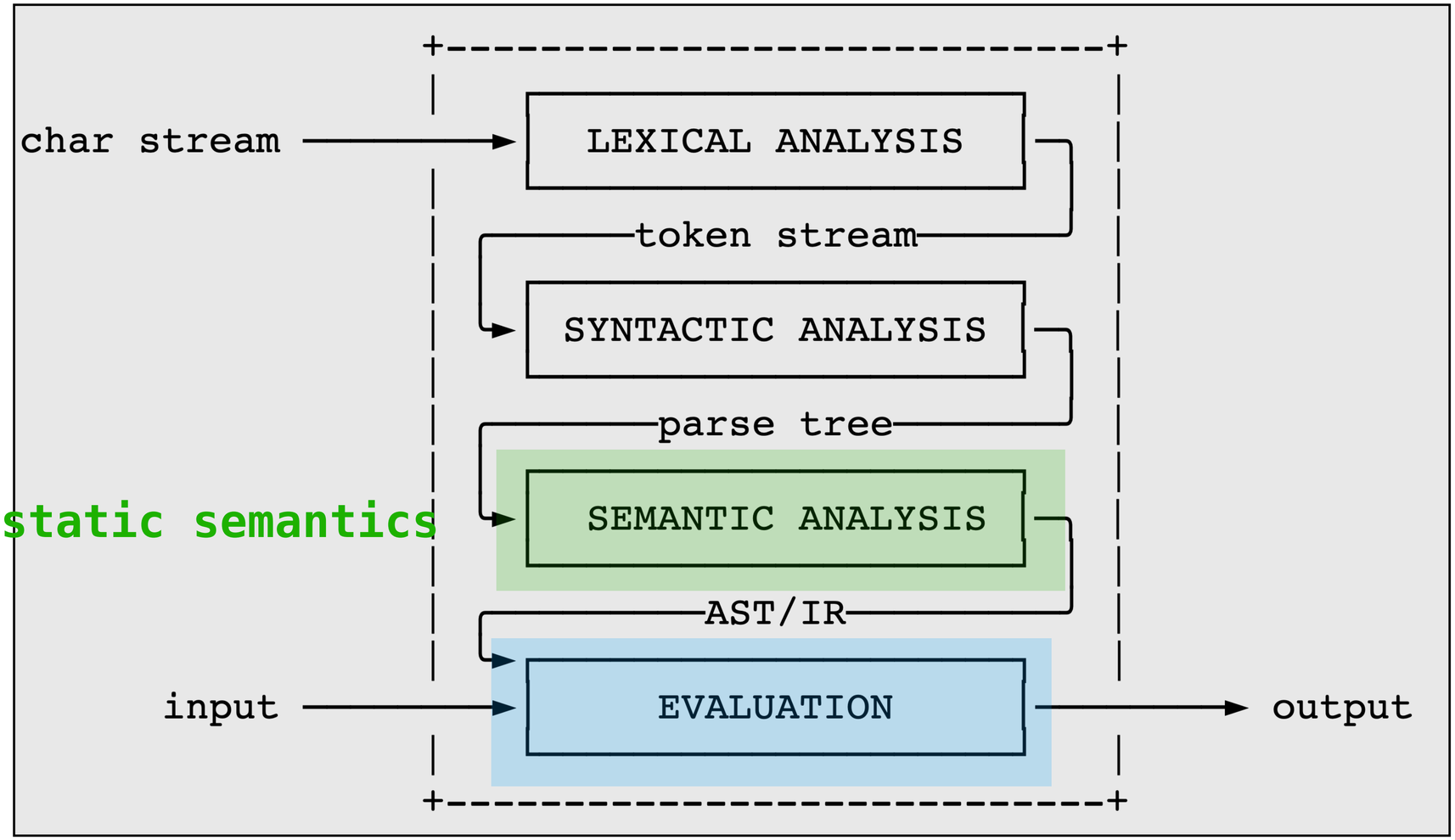


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dynamic semantics (this week)

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Small-Step Operational Semantics

Small-Step Semantic Judgment

$$(S, p) \longrightarrow (S', p')$$

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Notation. We write $e \longrightarrow e'$ to mean e reduces to e' in a single step

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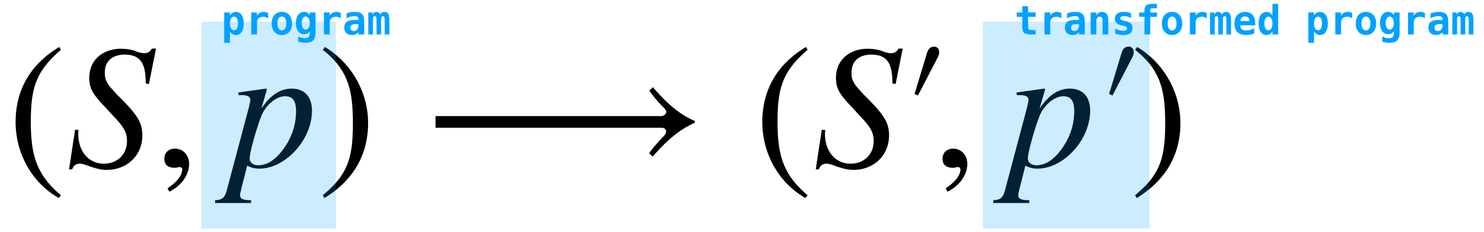
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We define small-step semantics on a **configuration,** which is a program together with some stateful information

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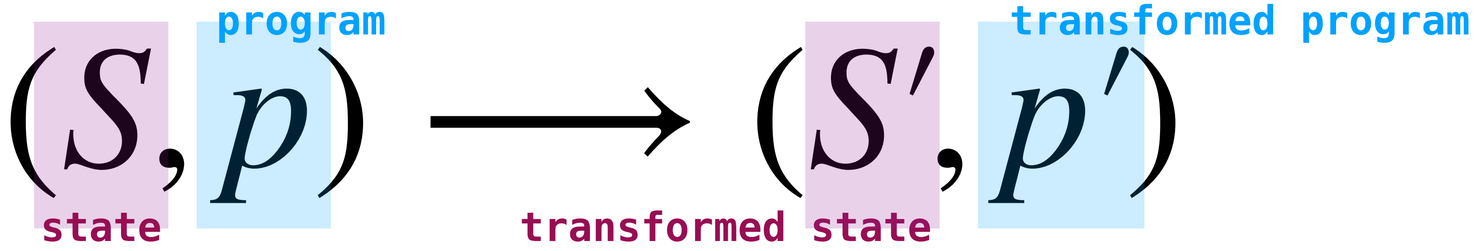


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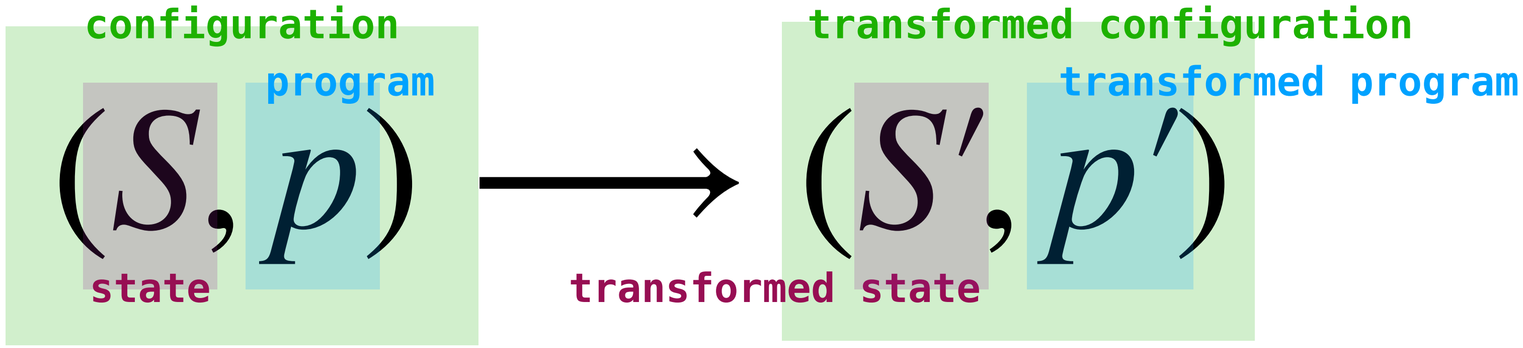


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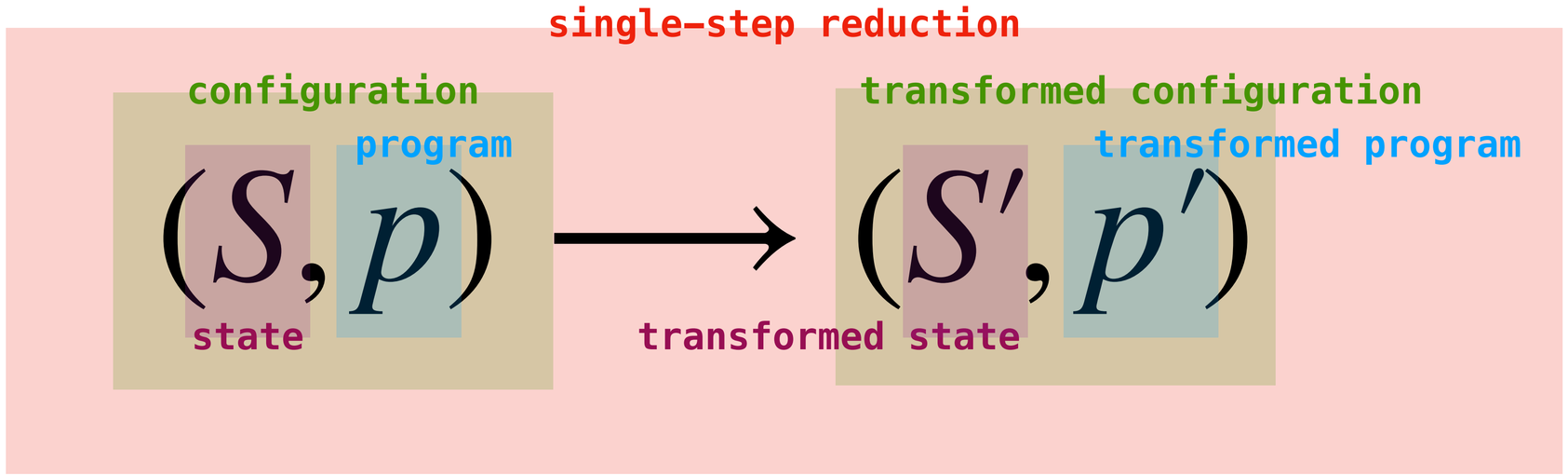


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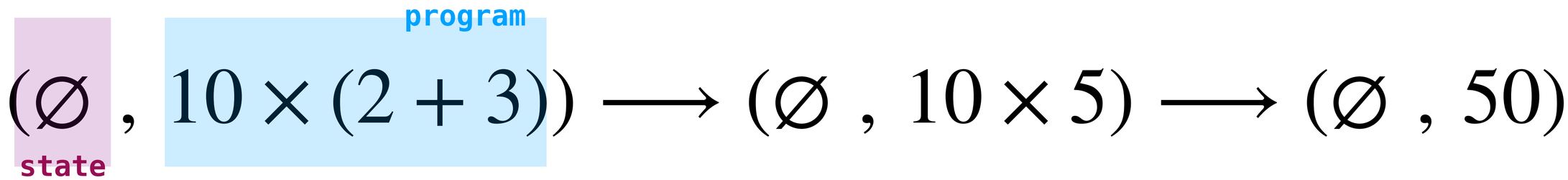


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We define small-step semantics on a **configuration,** which is a program together with some stateful information

Example: Arithmetic Expressions



State: none

Program: Arithmetic expression

Example: OCaml

program

state

$(\varnothing,$ `let x = 3 in if x > 10 then 4 else 5` $)$

$\longrightarrow (\varnothing,$ `if 3 > 10 then 4 else 5` $)$

$\longrightarrow (\varnothing,$ `if false then 4 else 5` $)$

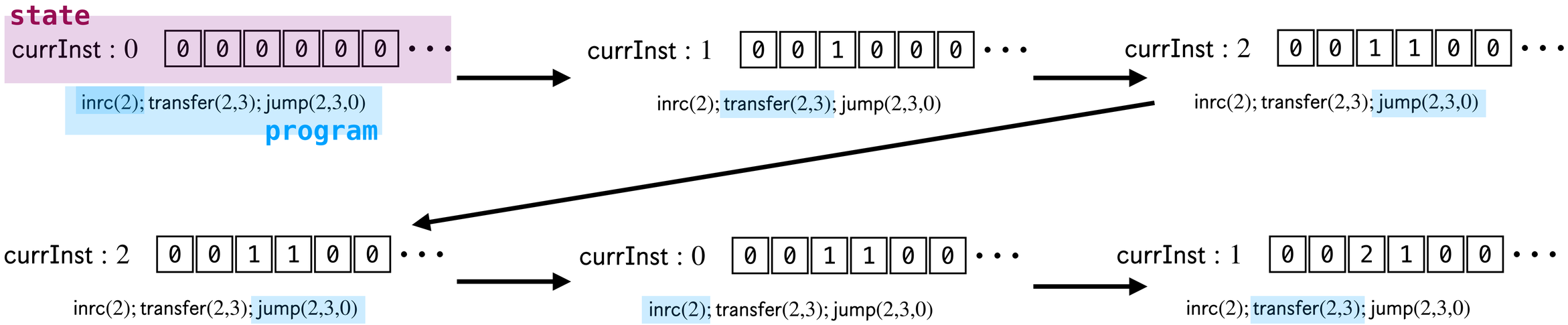
$\longrightarrow (\varnothing,$ `5` $)$

State: None

Program: OCaml expression

For purely functional languages
there is no state

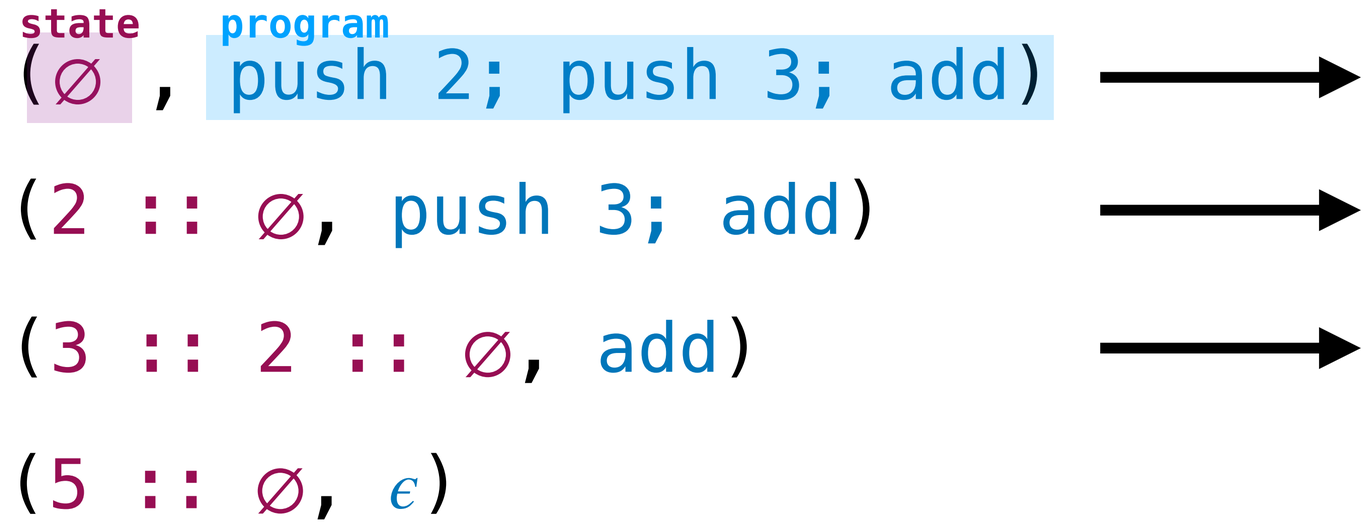
Example: Register Machines



State: (current instruction pointer) +
(collection of number registers)

Program: Sequence of commands

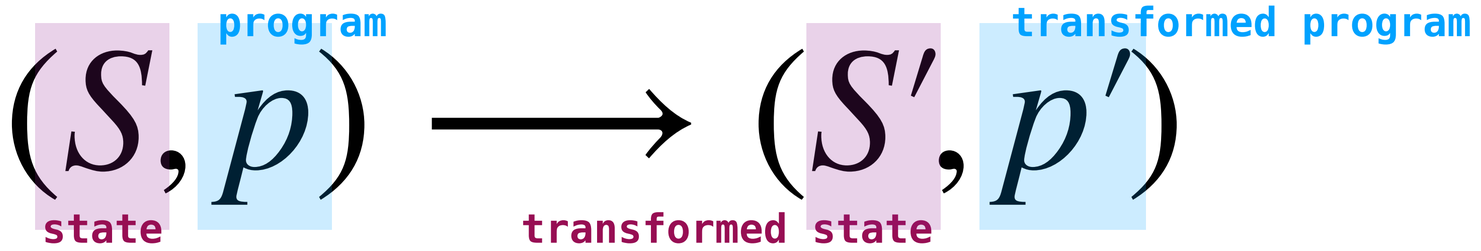
Example: Stack Language



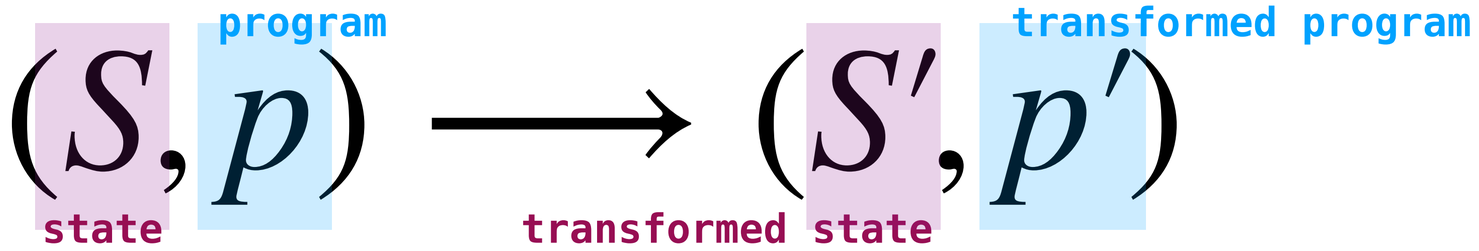
State: stack (i.e., list) of values

Program: sequence of commands for manipulating the stack

Small-Step Semantics

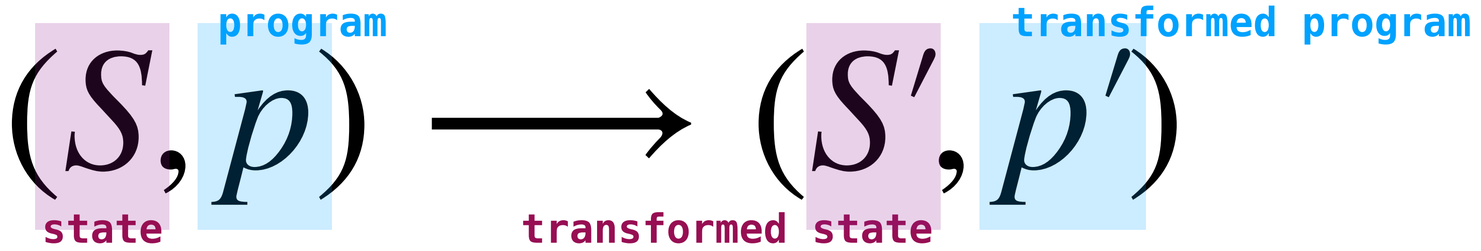


Small-Step Semantics



Defining a small-step semantics means three things:

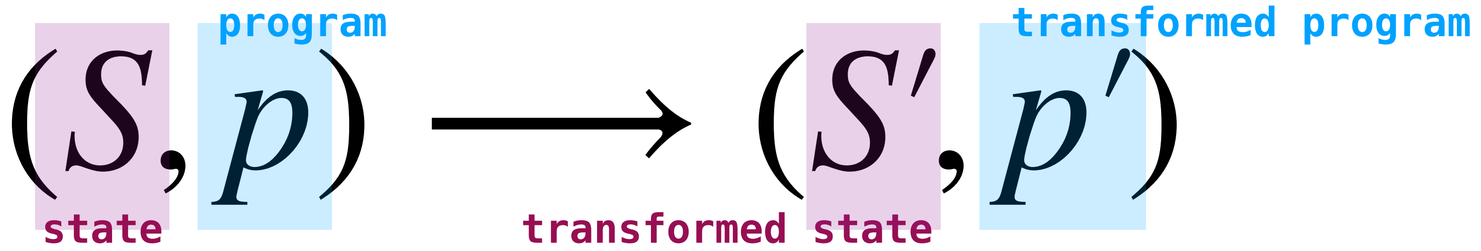
Small-Step Semantics



Defining a small-step semantics means three things:

» What kind of **state**?

Small-Step Semantics

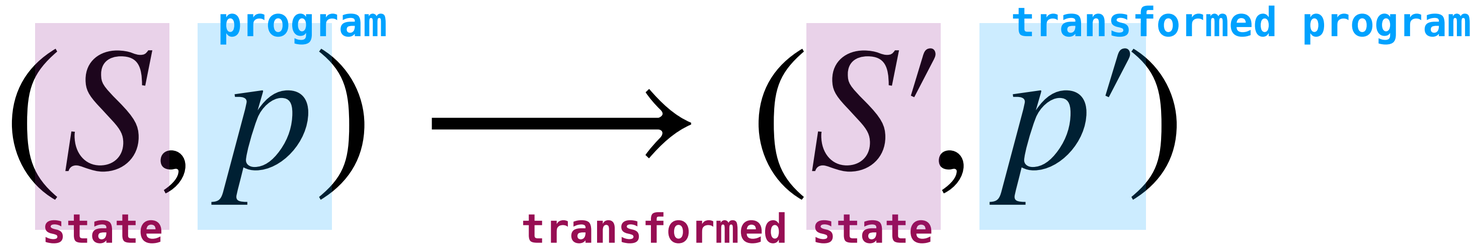


Defining a small-step semantics means three things:

» What kind of **state**?

» What **rules** for transforming configurations?

Small-Step Semantics



Defining a small-step semantics means three things:

- What kind of **state**?
- What **rules** for transforming configurations?
- What are the **successful** configurations?

Example

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

(eq (add 1 2) (add 2 3))

Example

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<expr> ::= ( <op> <expr> <expr> )
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$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left}$$

$$\frac{e_2 \longrightarrow e_2'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1\ \texttt{is a number} \qquad n_2\ \texttt{is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

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<expr> ::= ( <op> <expr> <expr> )
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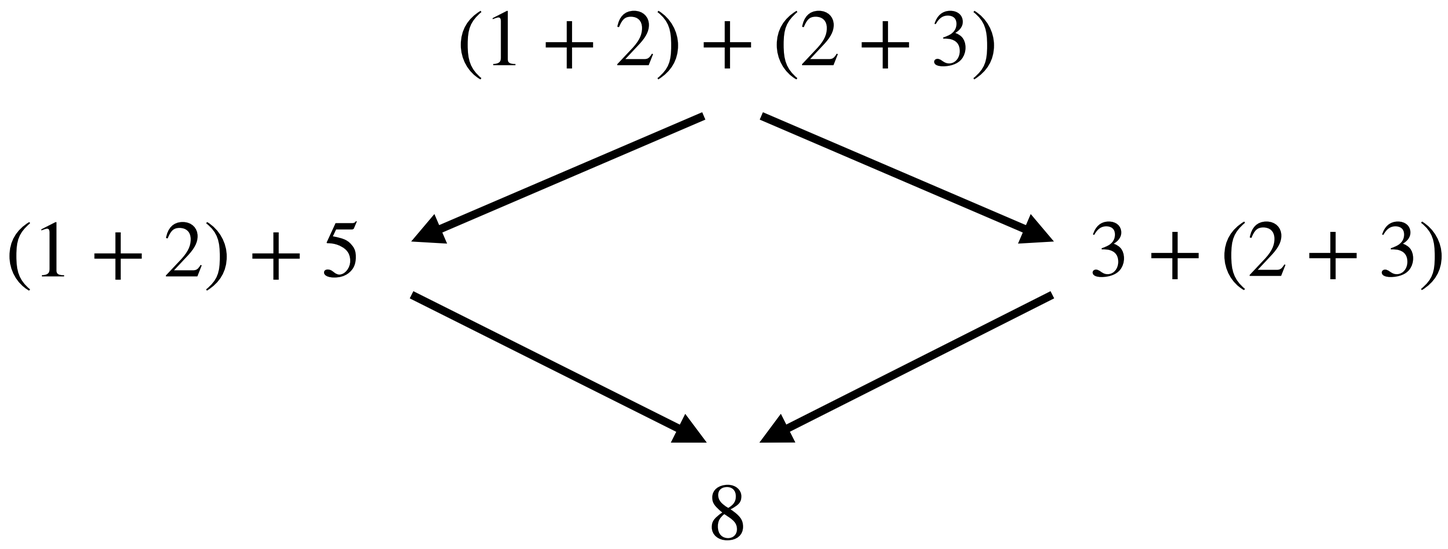
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Nondeterminism



There may be multiple choices of reductions

We should design our rules to *avoid* this

Nondeterminism

$$\frac{\text{add } 1\ 2 \longrightarrow 3}{(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow (\text{add } 3\ (\text{add } 2\ 3))}\ \texttt{add-left}$$

$$\frac{\text{add } 2\ 3 \longrightarrow 5}{(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow (\text{add } (\text{add } 1\ 2)\ 5)}\ \texttt{add-right}$$

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There are two reductions from (add (add 1 2) (add 2 3)) in our current rule set

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There are two reductions from (add (add 1 2) (add 2 3)) in our current rule set

We can avoid this by *breaking symmetry*

Example: Addition

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left}$$

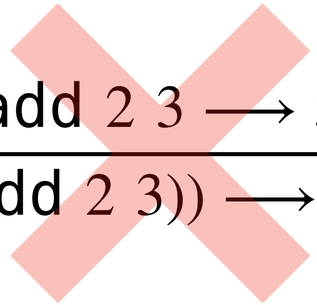
side condition

$$\frac{v\ \texttt{is a number} \qquad e_2 \longrightarrow e_2'}{(\texttt{add}\ v\ e_2) \longrightarrow (\texttt{add}\ v\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1\ \texttt{is a number} \qquad n_2\ \texttt{is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

Enforcing an Evaluation Strategy

$$\frac{\texttt{add}\ 1\ 2 \longrightarrow 3}{(\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3))}\ \texttt{add-left}$$



$$\frac{\texttt{add}\ 2\ 3 \longrightarrow 5}{(\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow (\texttt{add}\ (\texttt{add}\ 1\ 2)\ 5)}\ \texttt{add-right}$$

The new rule enforces that arguments of **add** are evaluated from left to right

Example

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

Example

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$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left}$$

$$\frac{n\ \texttt{is a number} \qquad e_2 \longrightarrow e_2'}{(\texttt{add}\ n\ e_2) \longrightarrow (\texttt{add}\ n\ e_2')}\ \texttt{add-right}$$

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Practice Problem

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<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
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<bool> ::= true | false
<int>  ::= ...
```

*Write down the reduction rules for **eq** (to the best of your ability) so that the left argument is evaluated before the right argument*

Answer

$$\frac{e_1 \to e_1'}{(\text{eq}\ e_1\ e_2) \to (\text{eq}\ e_1'\ e_2)}\ \text{eq-left}$$

$$\frac{v \text{ is value} \qquad e_2 \to e_2'}{(\text{eq}\ v\ e_2) \to (\text{eq}\ v\ e_2')}\ \text{eq-right}$$

$$\frac{v_1 \text{ is value} \quad v_2 \text{ is value} \quad v_1 = v_2}{(\text{eq}\ v_1\ v_2) \to \text{true}}\ \text{eq-true}$$

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

(?)

$$\frac{v_1 \text{ is value} \quad v_2 \text{ is value} \quad v_1 \neq v_2}{(\text{eq}\ v_1\ v_2) \to \text{false}}\ \text{eq-false}$$

(eq true 1) $\to$ false

(eq true 1) $\not\to$

Two Questions

Given a small-step semantics, we can ask:

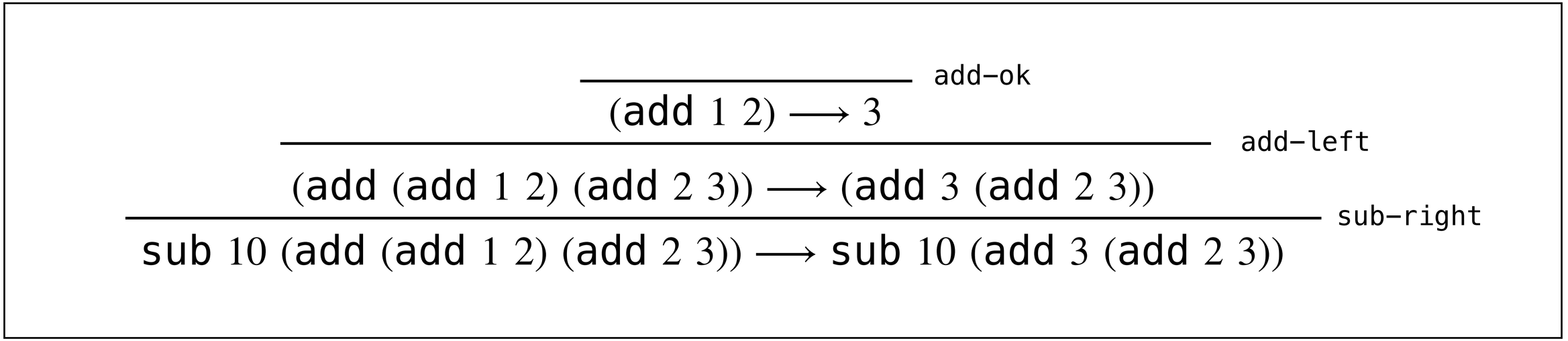
» Show that $C \longrightarrow C'$

» Given C, determine a configuration C' such that $C \longrightarrow C'$ (and show that it holds)

Recall: Derivations

$$\dfrac{\dfrac{\dfrac{}{(\text{add } 1\ 2) \longrightarrow 3}\ \texttt{add-ok}}{(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow (\text{add } 3\ (\text{add } 2\ 3))}\ \texttt{add-left}}{\text{sub } 10\ (\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow \text{sub } 10\ (\text{add } 3\ (\text{add } 2\ 3))}\ \texttt{sub-right}$$

Recall: Derivations



A **derivation** is a tree of reductions, gotten by applying reduction rules

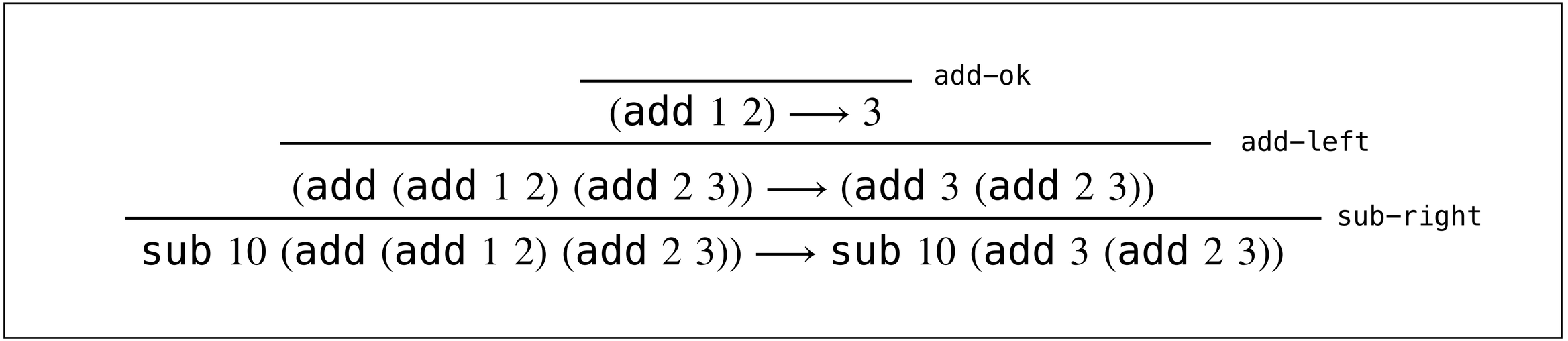
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A **derivation** is a tree of reductions, gotten by applying reduction rules

A derivation is a proof that the reduction step is valid in the operational semantics

Recall: Derivations



A **derivation** is a tree of reductions, gotten by applying reduction rules

A derivation is a proof that the reduction step is valid in the operational semantics

We've done this

Two Questions

Given a small-step semantics, we can ask:

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» Given C, determine a configuration C' such that $C \longrightarrow C'$ (and show that it holds)

Single-Step Evaluation

(sub 10 (add (add 1 2) (add 2 3))) ⟶ ???

Single-Step Evaluation

$$(\texttt{sub}\ 10\ (\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3))) \longrightarrow ???$$

The more "realistic" situation is to *figure out* what a program evaluates to in a single step

$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left} \qquad \frac{n\ \texttt{is a number} \quad e_2 \longrightarrow e_2'}{(\texttt{add}\ n\ e_2) \longrightarrow (\texttt{add}\ n\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1\ \texttt{is a number} \quad n_2\ \texttt{is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

Example

(sub 10 (add (add 1 2) (add 2 3))) ⟶ ???

$$\frac{}{(+\ 1\ 2) \longrightarrow 3}\ \texttt{add-ok}$$

$$\frac{(+\ 1\ 2) \longrightarrow 3}{(+\ (+\ 1\ 2)\ (+\ 2\ 3)) \longrightarrow (+\ 3\ (+\ 2\ 3))}\ \texttt{add-left}$$

$$\frac{(+\ (+\ 1\ 2)\ (+\ 2\ 3)) \longrightarrow (+\ 3\ (+\ 2\ 3))}{(-\ 10\ (+\ (+\ 1\ 2)\ (+\ 2\ 3))) \longrightarrow (-\ 10\ (+\ 3\ (+\ 2\ 3)))}\ \texttt{sub-right}$$

Practice Problem

$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left} \qquad \frac{e_2 \longrightarrow e_2'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1\ e_2')}\ \texttt{add-right}$$

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$$\frac{n_1\ \texttt{is a number} \quad n_2\ \texttt{is a number}}{(\texttt{sub}\ n_1\ n_2) \longrightarrow n_1 - n_2}\ \texttt{sub-ok}$$

$$(\texttt{sub}\ 10\ (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3))) \longrightarrow (\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5))$$

Give a derivation of the above reduction

Answer

$$(\texttt{sub}\ 10\ (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3))) \longrightarrow (\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5))$$

Multi-Step Reduction Relation

$$\frac{}{C \longrightarrow^\star C}\ \text{refl} \qquad \frac{C \longrightarrow C' \qquad C' \longrightarrow^\star D}{C \longrightarrow^\star D}\ \text{step}$$

Given any single-step reduction relation, we have a corresponding **multi-step reduction relation:**

» Every configuration reduces to itself **(reflexivity)**

» Every $\longrightarrow^\star$ reduction can be extended by a single step **(transitivity)**

Two Questions (Again)

Once we have an operational semantics, there are two questions we can ask (as PL designers and on the final exam):

» Show that $C \longrightarrow^\star C'$

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How To: Derivations of Multi-Step Reductions

$$\texttt{sub}\ 10\ (\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow^{\star} 2$$

want to show

» Derive all necessary single-step evaluations

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(we did this)

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$\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5) \longrightarrow \texttt{sub}\ 10\ 8$ (exercise)

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$\texttt{sub}\ 10\ 8 \longrightarrow 2$ (easy)

» Derive all necessary single-step evaluations

Traces

sub 10 (add (add 1 2) (add 2 3)) ⟶

sub 10 (add 3 (add 2 3)) ⟶

sub 10 (add 3 5) ⟶

sub 10 8 ⟶ 2

A **trace** is a sequence of (valid) single-step reductions

How To: Derivations of Multi-Step Reductions

$$\mathsf{sub}\ 10\ (\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3)) \longrightarrow^{\star} 2$$

» Derive all necessary single-step evaluations

» Combine them with the **transitivity rule**

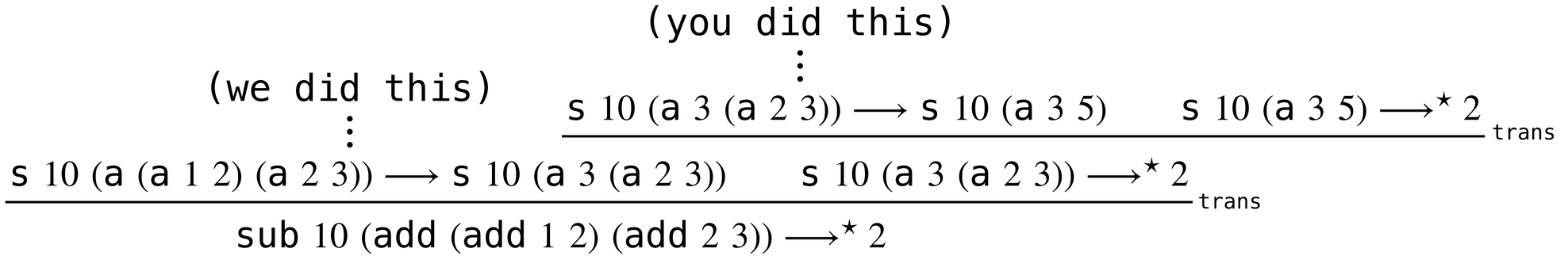
How To: Derivations of Multi-Step Reductions

$$\dfrac{\begin{array}{c}\texttt{(we did this)}\\ \vdots\\ \texttt{s}\ 10\ (\texttt{a}\ (\texttt{a}\ 1\ 2)\ (\texttt{a}\ 2\ 3)) \longrightarrow \texttt{s}\ 10\ (\texttt{a}\ 3\ (\texttt{a}\ 2\ 3)) \qquad \texttt{s}\ 10\ (\texttt{a}\ 3\ (\texttt{a}\ 2\ 3)) \longrightarrow^{\star} 2\end{array}}{\texttt{sub}\ 10\ (\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow^{\star} 2}\ \texttt{trans}$$

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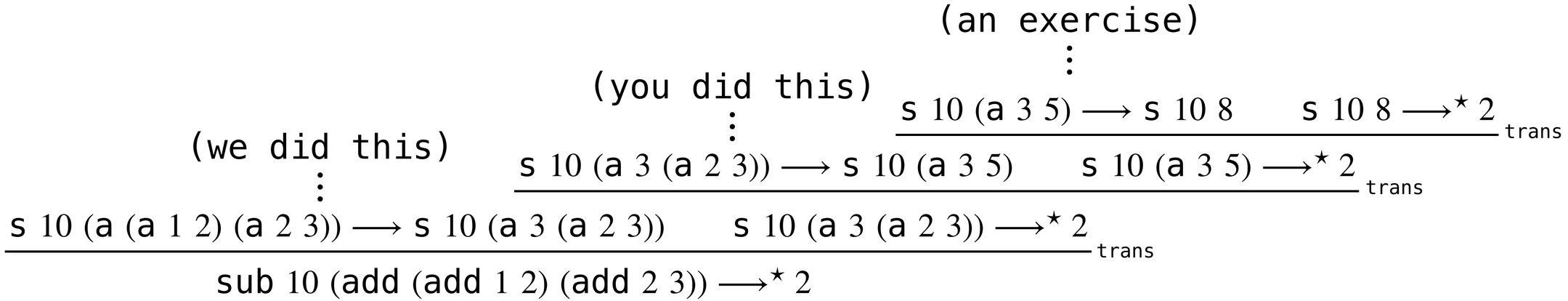
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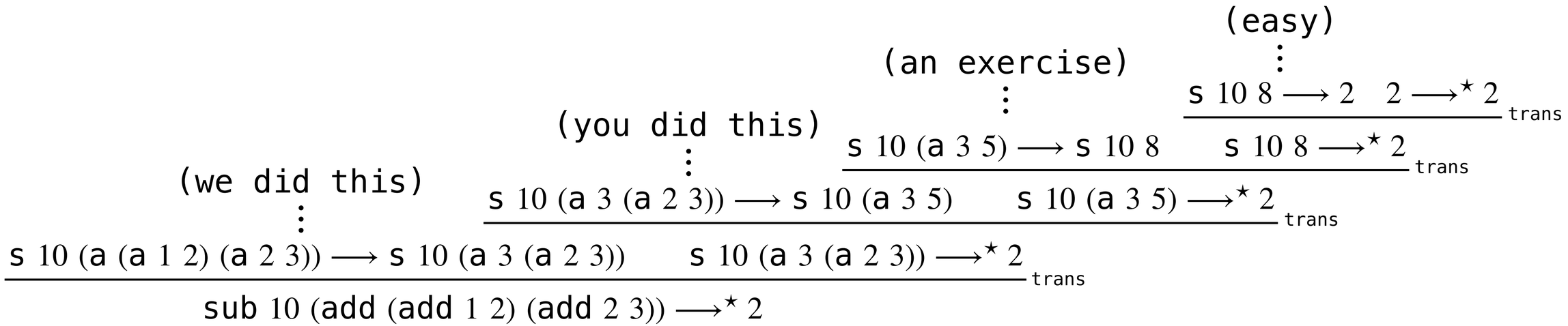
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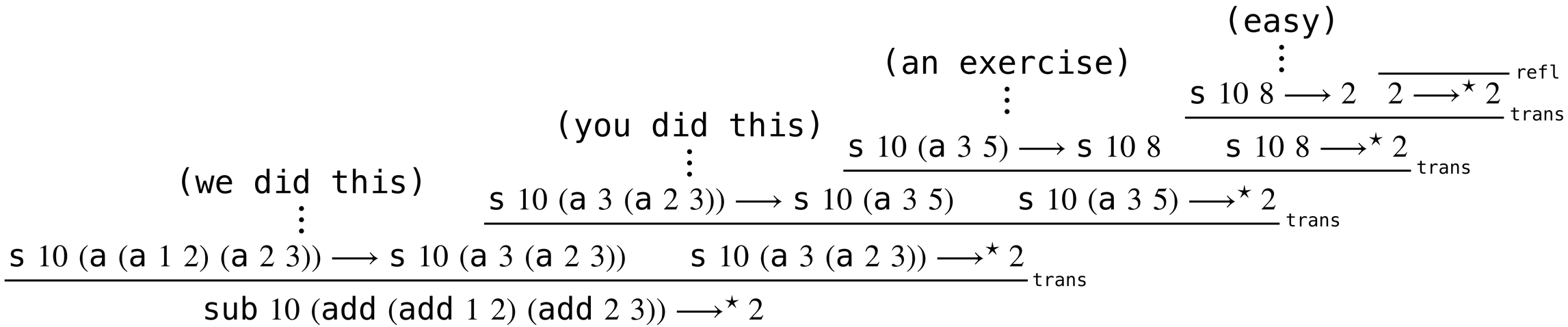
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Once we have an operational semantics, there are two questions we can ask (as PL designers and on the final exam):

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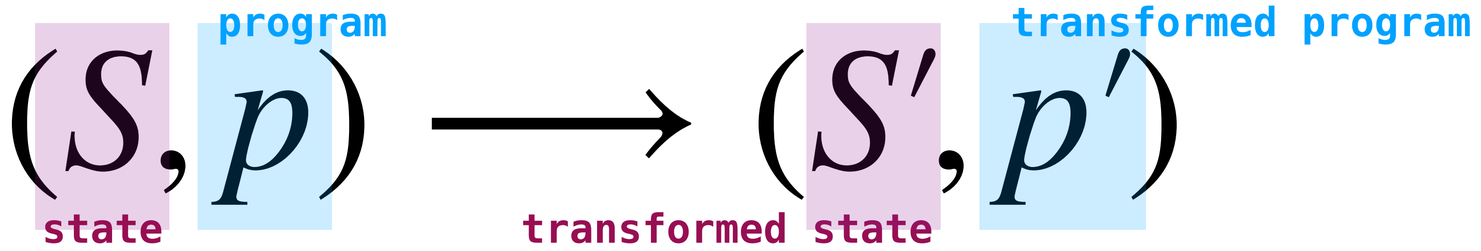
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Solve this single-step evaluation problem until you reach a configuration that cannot be further reduced

When are we "done" computing?

Recall: Small-Step Semantics



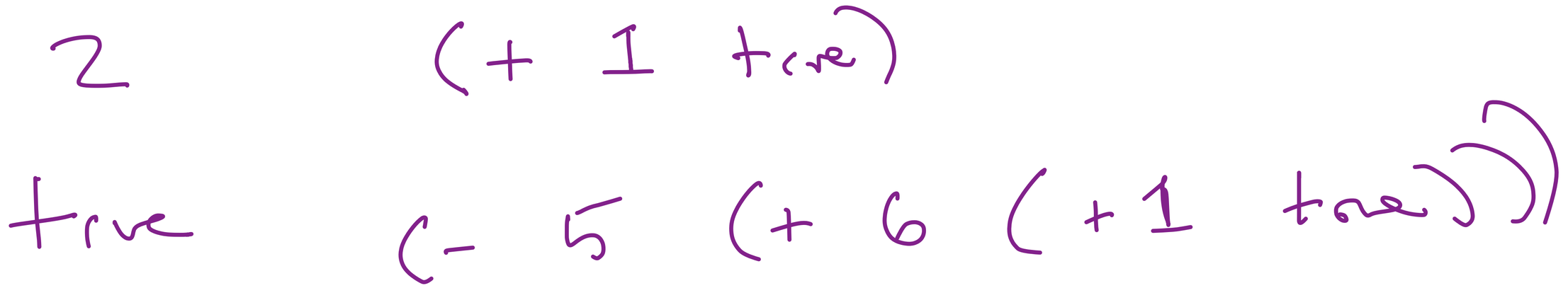
Defining a small-step semantics means three things:

» What kind of **state**?

» What **rules** for transforming configurations?

» What are the **successful** configurations?

Irreducible Configurations/Normal Forms



An **irreducible configuration** is one that cannot be further reduced

Not all irreducible configurations are "successes"

Final Configurations/Values

<value> ::= <int> | true | false

We need to say which configurations are "success"

We often do this by specifying a subset of the (abstract) syntax

An irreducible configuration that is not final is called **stuck**

Three End-States

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» **value:** we reach the value of our program

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```
(fun x -> x) (2 + 3) →* 5
```

Three End-States

» **value:** we reach the value of our program

$$\texttt{(fun x -> x) (2 + 3)} \rightarrow^\star \texttt{5}$$

» **stuck:** we reach an expression that cannot be reduced, but that is not a value

$$\texttt{y (fun x -> x)} \not\rightarrow$$

» **diverge:** the computation never reaches a point where the expression is not reducible

$$\texttt{(fun x -> x x) (fun x -> x x)} \rightarrow^\star \texttt{(fun x -> x x) (fun x -> x x)}$$

Big-Step Semantics

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We write $e \Downarrow v$ to mean that e evaluates to the value v

This is what we've been doing in this course so far

Example

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

$$\frac{n \text{ is a number}}{n \Downarrow n}\ \text{numEval}$$

$$\frac{e_1 \Downarrow v_1 \qquad e_2 \Downarrow v_2 \qquad v_1 \text{ is a number} \qquad v_2 \text{ is a number}}{(\text{add } e_1\ e_2) \Downarrow v_1 + v_2}\ \text{addEval}$$

$$\frac{e_1 \Downarrow v_1 \qquad e_2 \Downarrow v_2 \qquad v_1 \text{ is a number} \qquad v_2 \text{ is a number}}{(\text{sub } e_1\ e_2) \Downarrow v_1 - v_2}\ \text{subEval}$$

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we'll remove these side conditions once we have type-checking

Practice Problem

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

Write the rule for ***eq***

Answer

```
<expr> ::= ( <op> <expr> <expr> )
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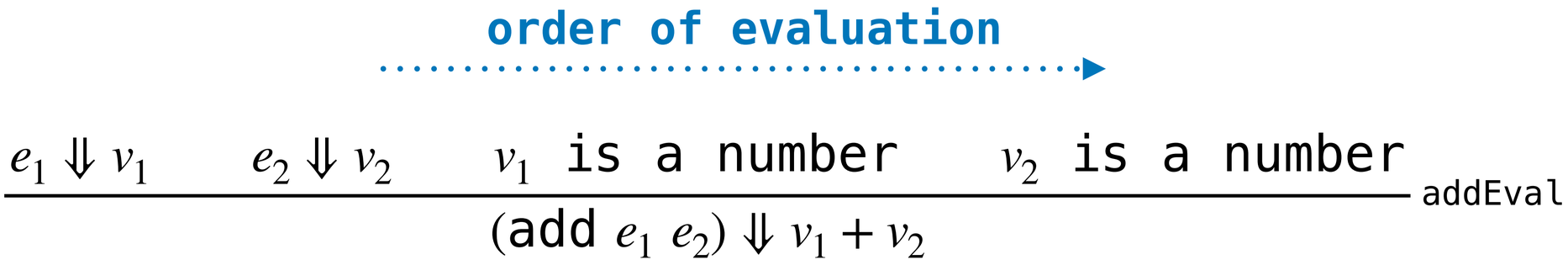
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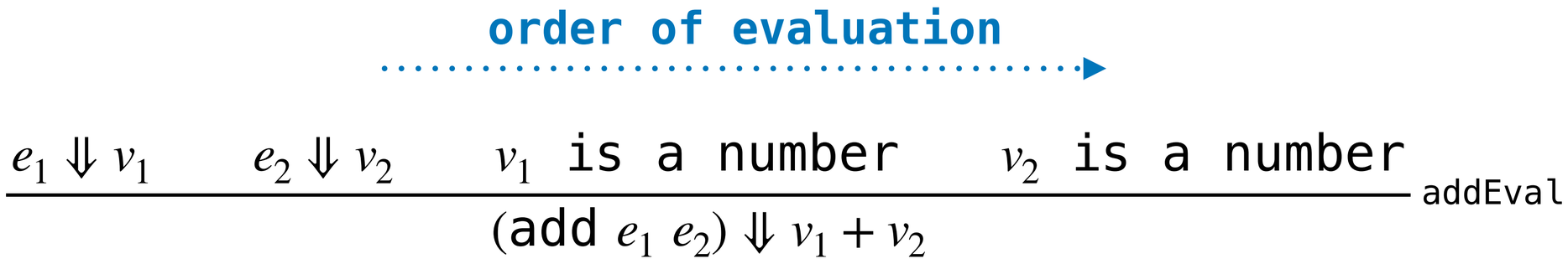
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Note: We can't always have both small-step and big-step!

Order of Evaluation

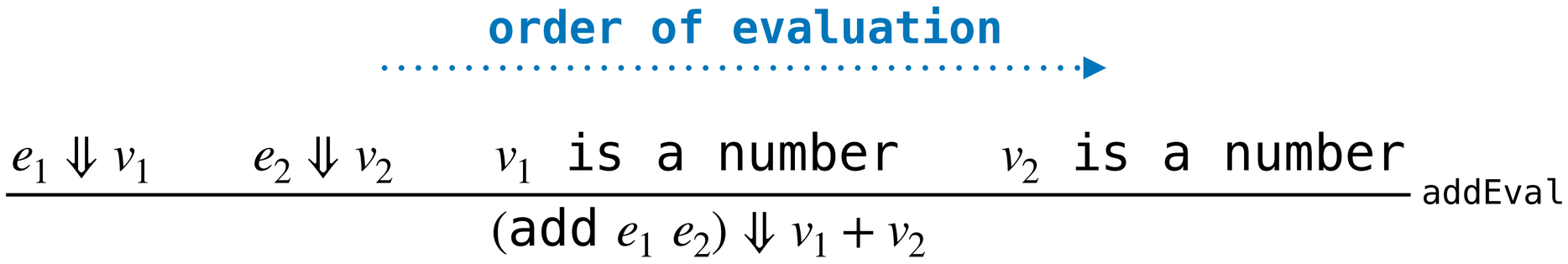


Order of Evaluation



With small-step semantics, we can choose the order of evaluations based on the rules

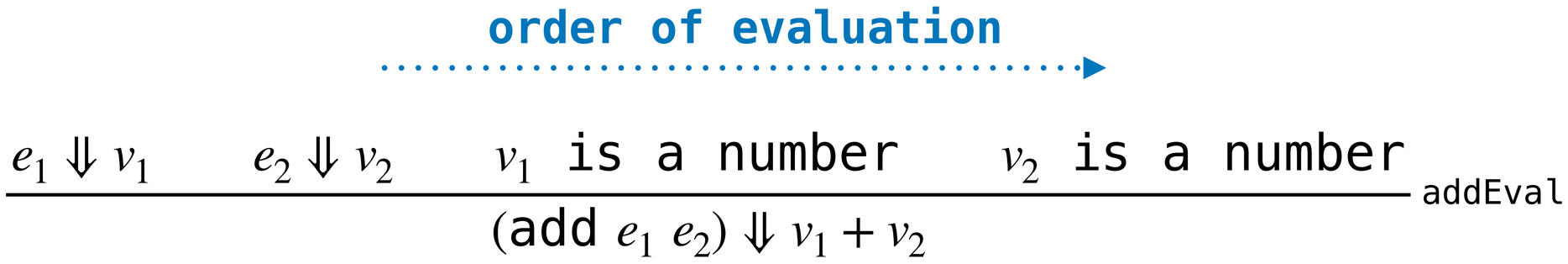
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With small-step semantics, we can choose the order of evaluations based on the rules

With big-step semantics, we can't because our relation only deals with the *final* value, nothing intermediate

We will take the order of evaluation to be from left to right

Summary

big-step

$$e \Downarrow v$$

e evaluates to v

single-step

$$e \longrightarrow e'$$

e reduces to e' in a single step

multi-step

$$e \longrightarrow^{\star} e'$$

e reduces to e' in many steps